

HELPS

*Helicopter Emergency Landing Place
Suggestion System*

System Design Document

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Terminology and Definitions

This section defines the key terms used throughout this System Design Document (SDD). These definitions establish a common understanding of the concepts, entities, and components that form the foundation of the **Helicopter Emergency Landing Place Suggestion System (HELPSs)**. Unless otherwise specified, the following terminology shall be used consistently throughout the remainder of this document.

- **Area of Interest (AOI):** A geographically bounded region selected for terrain analysis and knowledge generation. All satellite imagery, Digital Elevation Models (DEM), and auxiliary geospatial datasets processed by HELPSs correspond to the selected Area of Interest.
- **Terrain Knowledge Generation Pipeline:** The backend processing pipeline responsible for collecting geospatial data, identifying candidate terrain, extracting terrain characteristics, generating operational zones, and constructing the Terrain Knowledge Database. This pipeline is executed whenever a new geographical region is analysed or when existing terrain information requires updating.
- **Helicopter Landing Recommendation Pipeline:** The mission-time decision support pipeline responsible for retrieving candidate terrain from the Terrain Knowledge Database, evaluating it against helicopter-specific operational requirements, and generating ranked emergency landing recommendations.
- **Terrain Knowledge Database:** The centralized spatial database generated by the Terrain Knowledge Generation Pipeline. It stores all helicopter-independent terrain information, including terrain objects, operational zones, terrain characteristics, environmental attributes, and spatial relationships required during recommendation generation.
- **Terrain Object:** A contiguous region of candidate terrain identified after eliminating permanent obstacles such as buildings, dense vegetation, steep cliffs, water bodies, or other unsuitable regions. A Terrain Object represents a physically continuous area that is considered suitable for further analysis but is not yet guaranteed to satisfy helicopter landing requirements.
- **Candidate Terrain:** Terrain that satisfies the minimum criteria required for further analysis following obstacle removal. Candidate Terrain serves as the initial input to the terrain characterization and operational zone generation processes.
- **Terrain Characterization:** The process of extracting descriptive information from each Terrain Object. This includes geometric properties, terrain composition, surface characteristics, vegetation density, accessibility, surrounding infrastructure, and other helicopter-independent attributes required for subsequent recommendation generation.
- **Operational Zone:** A contiguous region within a Terrain Object that satisfies a specified terrain slope threshold. Multiple Operational Zones may exist within a single Terrain Object, each representing a different slope constraint to accommodate helicopters with varying operational limitations.

- **Slope Profile:** The complete collection of Operational Zones generated for a Terrain Object across multiple slope thresholds. Instead of representing terrain using a single slope value, HELPSs preserves a multi-threshold slope representation to support helicopters with different permissible landing slopes.
- **Representative Point:** A reference point located within an Operational Zone that is maximally distant from its surrounding boundaries. HELPSs utilizes the Pole of Inaccessibility as the Representative Point because it provides the safest and most centrally located position for evaluating landing feasibility and clearance.
- **Pole of Inaccessibility:** The point inside a polygon that has the maximum possible distance from all polygon boundaries. This point represents the center of the largest empty circle that can be inscribed within the polygon and is used by HELPSs as the reference location for landing footprint validation.
- **Landing Footprint:** The physical area required for a helicopter to safely occupy an Operational Zone, including the rotor diameter and the necessary operational safety clearance. Landing Footprint validation is performed during the Helicopter Landing Recommendation Pipeline and depends on helicopter-specific operational characteristics.
- **Environmental Context:** The collection of environmental attributes surrounding an Operational Zone, including accessibility, nearby infrastructure, surrounding obstacles, vegetation, water bodies, road connectivity, terrain openness, and other operationally relevant information used during recommendation ranking.
- **Approach Corridor:** The surrounding airspace through which a helicopter approaches or departs an Operational Zone. In Version 1 of HELPSs, the suitability of the approach corridor is represented using an Approach Corridor Score derived from surrounding terrain openness and nearby obstacle information.
- **Approach Corridor Score:** A numerical measure representing the relative openness and accessibility of an Operational Zone based on its surrounding environment. The score provides an estimate of how favorable the surrounding terrain is for helicopter approach and departure operations and serves as one of the factors used during recommendation ranking.
- **Mission Context:** The operational objective associated with a helicopter mission. Examples include emergency engine failure, fuel emergency, medical evacuation, disaster response, and search-and-rescue operations. Mission Context influences the weighting of ranking parameters during recommendation generation.
- **Recommendation Score:** A composite suitability score assigned to every candidate Operational Zone after evaluating helicopter compatibility, environmental conditions, landing feasibility, accessibility, mission context, and other operational constraints. This score is used to rank the final emergency landing recommendations presented to the pilot.

Chapter 1: Problem Understanding

1.1 Background

Helicopters play a critical role in **search and rescue, emergency medical services, disaster response, military operations, border surveillance, and offshore logistics** due to their ability to **take off and land vertically**, hover, and operate in environments inaccessible to conventional fixed-wing aircraft. These missions frequently require helicopters to operate over **dense forests, mountainous terrain, deserts, coastal regions, and other remote areas**, where dedicated landing infrastructure is often unavailable.

During emergency situations such as **engine failures, mechanical malfunctions, fuel emergencies, or adverse weather conditions**, pilots must rapidly identify a **safe landing location** while simultaneously maintaining control of the aircraft. This decision requires evaluating multiple terrain characteristics, including **terrain slope, surface conditions, surrounding obstacles, available landing clearance, and accessibility**, all within a limited time frame and under significant operational stress.

The increasing availability of **high-resolution satellite imagery, Digital Elevation Models (DEM), Geographic Information System (GIS) data, and geospatial analysis techniques** has made large-scale terrain understanding feasible. These technologies provide the foundation for developing **intelligent decision-support systems** capable of analyzing terrain, identifying potential emergency landing opportunities, and assisting pilots with **objective, data-driven recommendations**, while preserving the pilot's role as the final decision-maker.

1.2 Problem Statement

Helicopter emergency landing is a **time-critical decision-making problem** where pilots must rapidly identify a **safe and operationally feasible landing location** while simultaneously managing the aircraft under emergency conditions. Unlike fixed-wing aircraft, helicopters frequently operate in **remote and unprepared environments** where dedicated landing infrastructure is unavailable.

Identifying a suitable landing location involves evaluating multiple terrain characteristics, including **terrain slope, surface type, surrounding obstacles, vegetation density, water bodies, landing clearance, and accessibility**. Performing this assessment manually during an emergency significantly increases the **pilot's cognitive workload** and the likelihood of overlooking suitable landing opportunities.

Although modern navigation systems provide **positioning, mapping, and terrain visualization**, they **do not automatically evaluate terrain suitability** for helicopter emergency landings. The responsibility of interpreting the surrounding terrain and selecting a landing location remains entirely with the pilot.

Additionally, **different helicopters have different operational constraints**, such as **rotor diameter, landing footprint, allowable slope, and surface limitations**. Building separate terrain

databases for every helicopter type is **computationally inefficient, difficult to maintain, and not scalable**.

To address this challenge, **HELPSs** proposes a **helicopter-independent terrain knowledge framework** that transforms raw geospatial data into structured terrain information. Instead of directly searching for landing locations for a specific helicopter, the system first **understands and characterizes the terrain**, and later evaluates it using **helicopter-specific operational requirements** during mission execution.

This separation enables a **scalable, reusable, and explainable decision-support system** that assists pilots by **reducing search time, providing objective terrain analysis, and presenting ranked landing recommendations**, while ensuring that **the pilot remains the final decision-maker**.

1.3 Challenges in Helicopter Emergency Landing

Emergency helicopter landing is a **high-pressure, time-critical operation** in which pilots must simultaneously control the aircraft while identifying a safe landing location. Unlike planned landings, emergency situations provide **limited decision time**, often requiring rapid assessment of unfamiliar terrain under stressful conditions.

The complexity of this process arises from the need to evaluate multiple terrain characteristics simultaneously. A seemingly open area may contain **steep slopes, dense vegetation, water bodies, power lines, surrounding obstacles, or insufficient landing clearance**, making it unsuitable for helicopter operations. Furthermore, different helicopters have **different operational limitations**, requiring the landing location to satisfy aircraft-specific constraints such as allowable slope, landing footprint, and safety clearance.

The challenge therefore extends beyond locating an open area. It requires **rapid identification, evaluation, and comparison of multiple terrain regions** while ensuring that the selected location satisfies both **terrain safety** and **helicopter operational requirements** under severe time constraints.

1.4 Limitations of Existing Navigation and Landing Systems

Modern aviation navigation systems provide **accurate positioning, route planning, terrain visualization, and obstacle awareness**, significantly improving flight safety. However, these systems are **not specifically designed to recommend emergency landing locations** for helicopters.

Existing systems primarily present **geographical information**, leaving the pilot responsible for interpreting terrain characteristics and determining whether a location is suitable for landing. They do not automatically analyse **terrain slope, landing clearance, surface conditions, surrounding obstacles, environmental accessibility, and helicopter-specific operational constraints** as a unified decision-support process.

Additionally, most available systems are designed for **navigation rather than terrain understanding**. They do not construct a reusable knowledge base of terrain that can be

evaluated for different helicopter platforms and mission requirements. Consequently, pilots must continue to perform complex terrain assessment manually during emergency situations, increasing cognitive workload when rapid decision-making is most critical.

1.5 Importance of Terrain Understanding

The suitability of an emergency landing location is determined by the **characteristics of the terrain** rather than its visual appearance. An open area may still be unsuitable due to **steep slopes, dense vegetation, surrounding obstacles, poor surface conditions, or insufficient landing clearance**. Therefore, identifying a safe landing location requires a comprehensive understanding of the terrain rather than simply locating empty space.

Terrain understanding involves analyzing and characterizing geographical regions using information such as **terrain geometry, slope, surface type, vegetation, water bodies, environmental context, and accessibility**. By transforming raw geospatial data into structured terrain knowledge, potential landing locations can be systematically identified and evaluated before being assessed against helicopter-specific operational requirements.

A terrain-centric approach enables the creation of a **reusable and helicopter-independent knowledge base**, allowing the same terrain information to support multiple helicopter platforms and mission scenarios. This significantly reduces repeated computation while providing a scalable foundation for intelligent emergency landing recommendation systems.

1.6 Project Objectives

The primary objective of **HELPSs (Helicopter Emergency Landing Place Suggestion System)** is to develop an **AI-assisted terrain knowledge and decision-support system** capable of identifying, characterizing, evaluating, and recommending potential emergency landing locations for helicopters.

The specific objectives of the proposed system are:

- **Develop a Terrain Knowledge Generation Pipeline** capable of analyzing satellite imagery, Digital Elevation Models (DEM), and GIS datasets to identify candidate terrain.
- **Generate a helicopter-independent terrain knowledge database** by extracting geometric, environmental, and operational characteristics of each terrain object.
- **Create operational zones** based on multi-threshold terrain slope analysis while preserving terrain information for different helicopter capabilities.
- **Develop a Helicopter Landing Recommendation Pipeline** that evaluates terrain knowledge using helicopter-specific operational constraints such as landing footprint, allowable slope, and mission requirements.
- **Provide explainable and ranked landing recommendations** supported by terrain characteristics, environmental analysis, and operational evidence rather than blind suggestions.

- **Assist pilots in reducing decision time and cognitive workload** during emergency situations while ensuring that **the pilot remains the final decision-maker**.

1.7 Project Scope

The scope of **HELPSs Version 1** is limited to the development of a **terrain knowledge generation and helicopter landing recommendation system** for emergency landing scenarios using publicly available geospatial data.

The system focuses on:

- **Generating terrain knowledge** from satellite imagery, Digital Elevation Models (DEM), and GIS datasets.
- **Identifying candidate terrain** by eliminating regions containing permanent obstacles.
- **Characterizing terrain** using geometric, surface, environmental, and accessibility features.
- **Generating operational zones** through multi-threshold slope analysis.
- **Constructing a reusable, helicopter-independent Terrain Knowledge Database.**
- **Evaluating terrain** using helicopter-specific operational constraints to produce ranked emergency landing recommendations.
- Providing an **offline, locally deployable decision-support system** that preserves organizational confidentiality without relying on cloud-based processing.

HELPSs excludes **predefined restricted areas**, such as military installations, airport runways, border-sensitive regions, government-protected facilities, and other legally prohibited zones, from terrain analysis and recommendation generation.

The current scope does **not** include autonomous flight control, automatic landing, real-time satellite imagery acquisition, or direct integration with onboard avionics.

1.8 Assumptions

The proposed system is developed based on the following assumptions:

- **Satellite imagery, DEM, and GIS datasets** are available for the selected Area of Interest (AOI).
- Terrain conditions are assumed to remain **relatively static** between successive Terrain Knowledge Generation cycles.
- Helicopter operational specifications, such as **rotor diameter, landing footprint, allowable slope, and safety clearance**, are available during the recommendation phase.
- Weather and wind information can be obtained from reliable external sources when generating recommendations.
- The system operates as a **decision-support tool**, and the **pilot retains full authority** over the final landing decision.

- Publicly available geospatial datasets provide sufficient resolution for identifying candidate terrain suitable for Version 1.

1.9 Limitations

Although HELPSs provides a structured framework for terrain understanding and emergency landing recommendation, Version 1 has several limitations.

The system primarily relies on **publicly available geospatial datasets**, whose spatial resolution and update frequency may limit terrain accuracy. Dynamic environmental conditions such as **temporary obstacles, moving vehicles, wildlife, recent construction, or vegetation changes** are not considered unless reflected in updated datasets.

The current implementation provides an **approximate approach corridor assessment** based on surrounding terrain openness and obstacle information rather than detailed helicopter flight dynamics. Similarly, weather and wind conditions are incorporated during the recommendation phase but are **not integrated into terrain knowledge generation**.

HELPSs is designed as a **decision-support system** and **does not replace pilot judgment**. The recommendations generated by the system are intended to assist situational awareness and reduce decision-making time, while the pilot remains responsible for evaluating operational safety and selecting the final landing location.

1.10 Chapter Summary

The need for an intelligent emergency landing assistance system arises from the **time-critical nature of helicopter emergency operations** and the complexity of evaluating terrain under high cognitive workload. Existing navigation systems primarily provide situational awareness but do not perform **comprehensive terrain suitability analysis** for emergency landings.

A safe emergency landing depends on **multiple terrain characteristics**, including geometry, slope, surface conditions, surrounding obstacles, environmental accessibility, and helicopter-specific operational constraints. This highlights the importance of transforming raw geospatial data into **structured terrain knowledge** that can be systematically evaluated during mission execution.

HELPSs addresses this need by adopting a **terrain-centric, helicopter-independent, and explainable decision-support approach**, establishing the foundation for the system architecture and design principles presented in the following chapters.

Chapter 2: Design Philosophy

2.1 Vision of HELPSs

The vision of **HELPSs (Helicopter Emergency Landing Place Suggestion System)** is to develop an **AI-assisted terrain knowledge and decision-support system** that enables rapid identification and evaluation of potential emergency landing locations for helicopters operating in challenging environments.

Rather than directly searching for landing sites, HELPSs aims to **understand and characterize the terrain** by transforming raw geospatial data into structured terrain knowledge. This knowledge can then be evaluated for different helicopter platforms based on their operational requirements, enabling the same terrain information to support multiple aircraft without repeated analysis.

HELPSs is designed as a **decision-support system**, not an autonomous decision-making system. The objective is to **reduce the pilot's cognitive workload** by narrowing the search space and presenting **ranked, explainable, and evidence-based landing recommendations**, while ensuring that the **pilot remains the final authority** in selecting the landing location. The long-term vision of HELPSs is to establish a **scalable and reusable terrain intelligence framework** that can be extended to different geographical regions, integrated with higher-resolution datasets, and adapted for future operational requirements without fundamentally changing the system architecture.

2.2 Design Principles

The architecture of HELPSs is guided by a set of fundamental design principles that ensure the system remains **modular, scalable, explainable, and adaptable**.

- **Terrain-First Architecture:** The system first develops a comprehensive understanding of the terrain before considering helicopter-specific operational constraints. This separation prevents repeated terrain analysis for different helicopter platforms.
- **Helicopter-Independent Knowledge Generation:** All terrain analysis is performed independently of any specific helicopter. Helicopter characteristics are considered only during the recommendation phase, allowing a single Terrain Knowledge Database to support multiple aircraft types.
- **Progressive Terrain Characterization:** Instead of immediately rejecting terrain based on helicopter limitations, HELPSs progressively extracts and stores terrain characteristics such as geometry, slope, surface composition, environmental context, and accessibility. This preserves valuable information for future evaluation.
- **Explainable Recommendations:** Every recommendation generated by HELPSs is accompanied by supporting evidence, including terrain characteristics, operational constraints, suitability scores, and environmental information. The system provides transparent reasoning rather than blind recommendations.

- **Pilot-Centric Decision Support:** HELPSs is designed to assist pilots, not replace them. The system provides objective terrain analysis while ensuring that the final operational decision always remains with the pilot.
- **Confidential and Local Deployment:** The recommendation engine is intended to operate entirely within a **local computing environment**, allowing organizations to perform mission planning and emergency landing analysis without transmitting sensitive operational data to external cloud services.
- **Scalable and Modular Design:** Each functional component of HELPSs is developed as an independent module with clearly defined inputs and outputs. This modular architecture simplifies maintenance, supports future enhancements, and enables integration of additional datasets or analytical capabilities without redesigning the complete system.
- HELPSs generates recommendations only from terrain that satisfies operational and regulatory constraints. Regions designated as restricted or prohibited for helicopter landing are excluded from analysis before terrain knowledge generation.

2.3 Terrain-First Philosophy

HELPSs adopts a **terrain-first approach**, where the system first analyzes and understands the terrain before considering any helicopter-specific constraints. By separating **terrain understanding** from **helicopter evaluation**, the same terrain knowledge can be reused for multiple helicopter platforms without repeated analysis, making the system scalable and efficient.

2.4 Helicopter-Independent Terrain Knowledge

Instead of generating landing locations for individual helicopters, HELPSs constructs a **helicopter-independent Terrain Knowledge Database** containing terrain geometry, surface characteristics, operational zones, and environmental information. During recommendation, this knowledge is evaluated using the operational constraints of the selected helicopter, enabling a single database to support multiple aircraft types.

2.5 Pilot-Centric Decision Support

HELPSs is designed to **assist, not replace, the pilot**. The system provides **ranked and explainable landing recommendations** supported by terrain analysis and operational evidence, while ensuring that **the pilot remains the final decision-maker** in all emergency landing scenarios.

2.6 Explainable Recommendation Strategy

HELPSs provides **explainable recommendations** rather than blind suggestions. Every recommended landing location is accompanied by supporting information such as **distance to the location, terrain characteristics, slope, surface type, accessibility, environmental context, and suitability score**, enabling pilots to understand **why** a location has been recommended.

2.7 Confidentiality and Local Processing

HELPSs is designed to operate as a **locally deployable decision-support system**, ensuring that sensitive mission information, helicopter specifications, and operational data remain within the organization's secure environment. Terrain knowledge is generated locally, and all recommendation computations are performed without relying on external cloud services, thereby preserving **confidentiality, security, and operational integrity**.

2.8 Adaptability to Different Geographic Regions

HELPSs is designed as a **geographically adaptable framework** rather than a region-specific solution. By utilizing satellite imagery, Digital Elevation Models (DEM), and GIS datasets, the system can generate terrain knowledge for **any Area of Interest (AOI)**. This enables the same architecture to be deployed across different terrains and geographical regions with minimal modifications, making the system scalable for future operational requirements.

2.9 Scalability Towards Real-Time Deployment

HELPSs is designed with a **modular and scalable architecture** that supports future enhancements without requiring significant redesign. Although Version 1 generates terrain knowledge using periodically updated geospatial datasets, the same architecture can be extended to incorporate **higher-resolution imagery, frequently updated terrain information, and real-time environmental data**. This design enables HELPSs to evolve from an offline terrain knowledge system into a **near real-time decision-support framework** as improved data sources become available.

2.10 Chapter Summary

The design philosophy of HELPSs is founded on the principle of **understanding terrain before evaluating helicopters**. By separating **terrain knowledge generation** from **helicopter-specific evaluation**, the system remains **modular, scalable, and reusable** across multiple helicopter platforms and operational scenarios.

HELPSs is designed as an **explainable, pilot-centric, and locally deployable decision-support system**, ensuring that recommendations are transparent, operationally meaningful, and generated without compromising organizational confidentiality. These design principles establish the foundation for the system architecture and implementation methodology presented in the subsequent chapters.

Chapter 3: Overall System Architecture

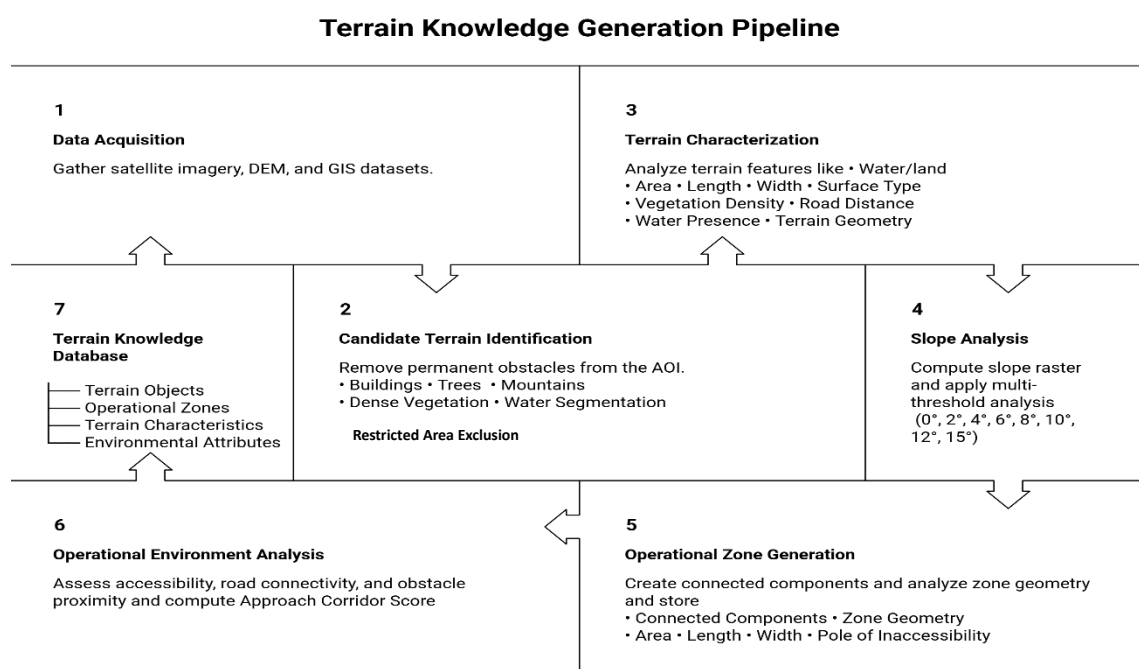
3.1 System Overview

HELPSs adopts a **modular two-stage architecture** that separates **terrain understanding** from **helicopter-specific landing evaluation**. This separation is one of the fundamental design principles of the proposed system, allowing terrain analysis to be performed independently of any helicopter platform while enabling the resulting terrain knowledge to be reused for different operational scenarios.

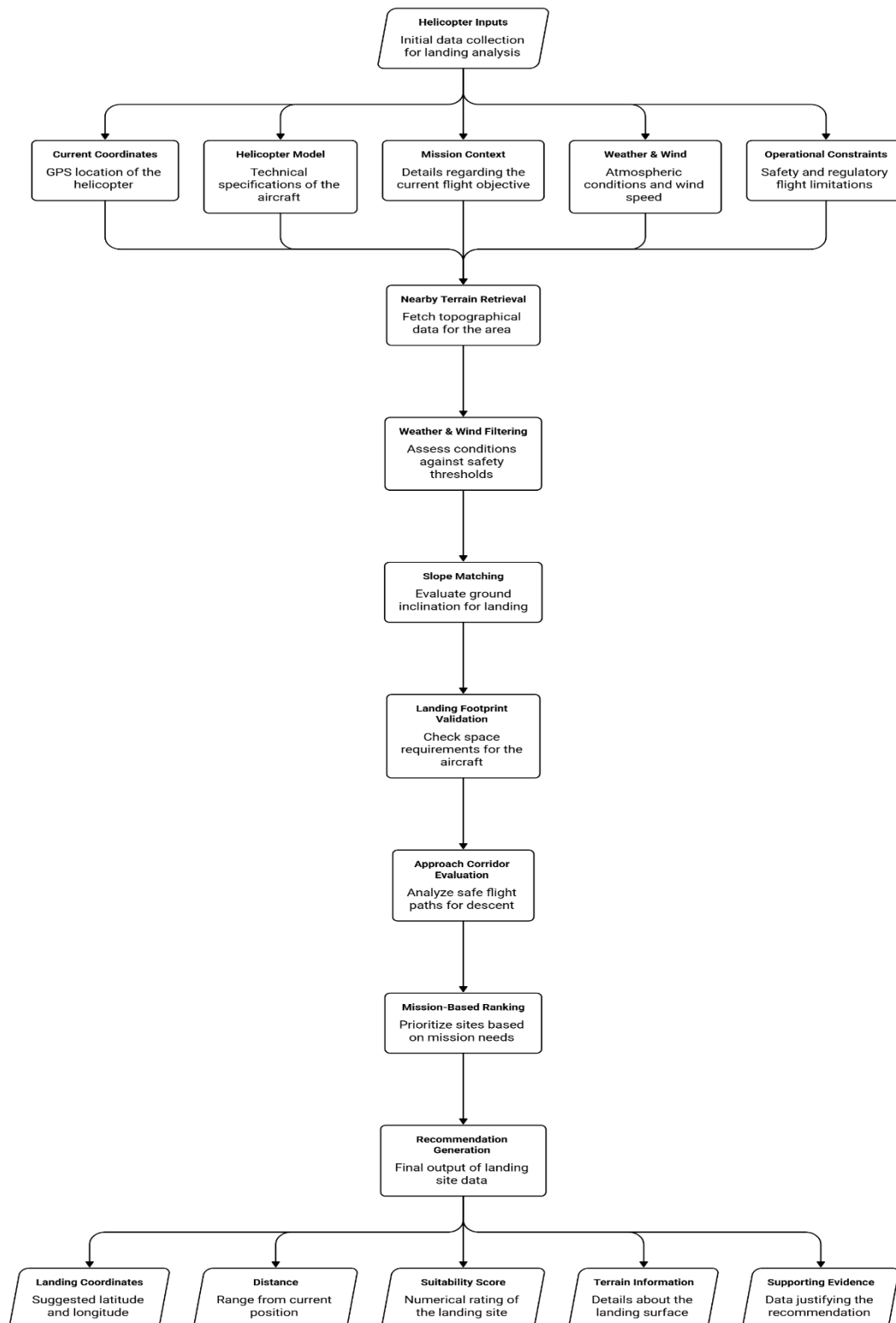
The overall architecture consists of two major processing pipelines: the **Terrain Knowledge Generation Pipeline** and the **Helicopter Landing Recommendation Pipeline**, interconnected through a centralized **Terrain Knowledge Database**. The Terrain Knowledge Generation Pipeline is responsible for analyzing raw geospatial data and constructing a comprehensive, helicopter-independent representation of the terrain. The Helicopter Landing Recommendation Pipeline subsequently utilizes this knowledge to evaluate terrain suitability based on helicopter-specific operational requirements and mission context.

By separating terrain analysis from operational evaluation, HELPSs avoids repeated geospatial computation for every helicopter platform while maintaining scalability, modularity, and adaptability. This architecture also enables the system to operate as a **locally deployable decision-support framework**, preserving organizational confidentiality while allowing future integration with improved datasets and analytical capabilities.

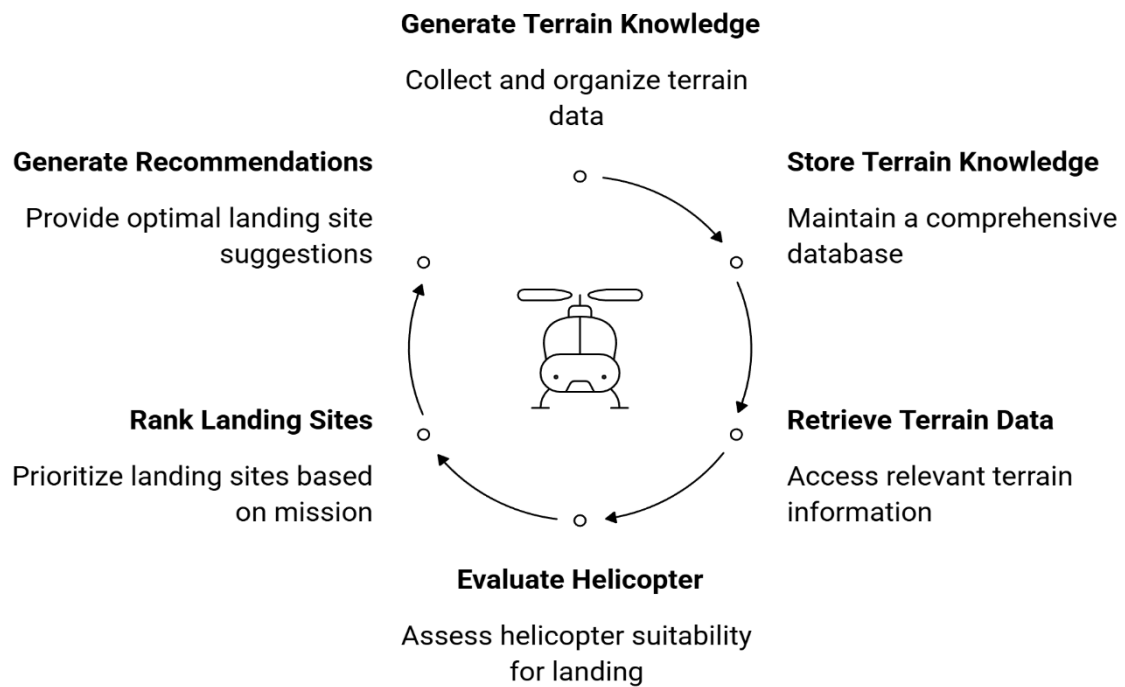
3.2 Terrain Knowledge Generation Pipeline



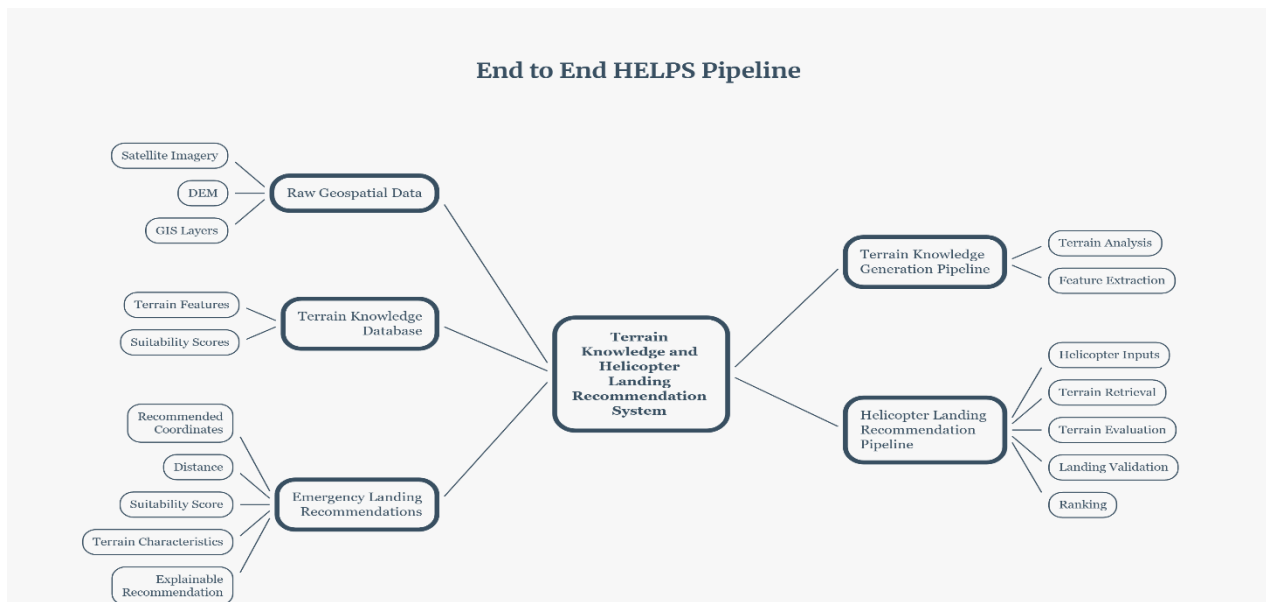
3.3 Helicopter Landing Recommendation Pipeline



3.4 Terrain Knowledge Integration



3.5 End-to-End Data Flow



Chapter 4: Terrain Knowledge Generation

Module 1: Data Acquisition

4.1.1 Overview

The **Data Acquisition** module serves as the foundation of the Terrain Knowledge Generation Pipeline. Its primary objective is to collect all geospatial datasets required to understand and characterize a selected **Area of Interest (AOI)**. Rather than relying on a single data source, HELPSs integrates multiple complementary datasets to capture the terrain's geometric, topographical, environmental, and infrastructural characteristics.

The acquired datasets include **satellite imagery**, **Digital Elevation Models (DEM)**, and **Geographic Information System (GIS)** layers. Together, these datasets provide the necessary information to identify candidate terrain, analyze elevation and slope, characterize surface properties, determine accessibility, and generate operational zones for emergency landing assessment.

Since terrain characteristics generally change slowly, data acquisition is performed **periodically** or whenever a new geographical region is selected for analysis. The collected datasets form the input for all subsequent modules within the Terrain Knowledge Generation Pipeline.

4.1.2Inputs

The Data Acquisition module requires the following inputs:

Input Dataset	Purpose
Area of Interest (AOI)	Defines the geographical region to be analyzed.
Satellite Imagery	Provides high-resolution visual information for identifying terrain features, vegetation, water bodies, and obstacles.
Digital Elevation Model (DEM)	Represents terrain elevation required for slope computation and terrain geometry analysis.
GIS Datasets	Provide vector information such as roads, rivers, administrative boundaries, and other geographic features.

4.1.3 Methodology

The Data Acquisition module is implemented through a sequence of standardized stages to ensure that all geospatial datasets required for terrain analysis are collected, processed, and organized consistently. Each stage prepares the necessary inputs for the subsequent modules while maintaining spatial consistency across all datasets.

Step 1: Define the Area of Interest (AOI)

Objective: Define the geographical region for which terrain knowledge will be generated. The AOI establishes the common spatial boundary for all datasets used throughout the Terrain Knowledge Generation Pipeline.

Implementation

- Select the geographical region to be analyzed.
- Represent the AOI using a **rectangular bounding box** defined by minimum and maximum latitude and longitude.
- Assign a unique AOI identifier.
- Store the corresponding Coordinate Reference System (CRS).
- Use the AOI as the reference boundary for acquiring all geospatial datasets.

Engineering Decision: A rectangular bounding box is selected for **Version 1** due to its simplicity, computational efficiency, and compatibility with most geospatial data providers. Support for polygon-based AOIs can be incorporated in future versions for more precise region selection.

Output

- AOI Definition
- Bounding Box Coordinates
- CRS Information

Step 2: Acquire Satellite Imagery

Objective: Acquire high-resolution satellite imagery of the selected AOI for identifying terrain features such as open land, vegetation, water bodies, and man-made structures.

Implementation

- Query the selected AOI from the satellite imagery provider.
- Retrieve the most recent cloud-free imagery.
- Clip the imagery to the AOI boundary.
- Store the processed imagery in **GeoTIFF** format.
- Record acquisition metadata for future reference.

Engineering Decision: **Sentinel-2 MSI** is selected for Version 1 because it provides free access, global coverage, regular updates, and adequate spatial resolution for large-scale terrain analysis.

Output

- Satellite Imagery
- Image Metadata

Step 3: Restricted Area Exclusion

Objective: Remove regions where helicopter landing is prohibited due to operational, legal, or security restrictions.

Implementation

- Load the predefined Restricted Area database.
- Identify intersections between Restricted Areas and Candidate Terrain.
- Remove restricted regions from further processing.
- Preserve only terrain that is legally and operationally permissible for analysis.

Typical Restricted Areas include:

- Airport runways
- Military installations
- International border-sensitive regions
- Government-protected facilities
- Critical infrastructure
- Permanently designated no-landing zones

Engineering Decision: Restricted Areas are excluded during terrain generation rather than during helicopter recommendation. This prevents prohibited locations from entering the Terrain Knowledge Database and ensures all stored terrain knowledge complies with operational restrictions.

Output: Legally compliant Candidate Terrain.

Step 4: Acquire Digital Elevation Model (DEM)

Objective: Acquire elevation data required for terrain geometry analysis and slope computation.

Implementation

- Download DEM tiles covering the AOI.
- Merge multiple tiles if necessary.
- Clip the DEM to the AOI boundary.
- Store the processed DEM in GeoTIFF format.
- Verify spatial alignment with the satellite imagery.

Engineering Decision: SRTM DEM (30 m) is selected due to its global availability, reliability, and suitability for the terrain-scale analysis required by HELPSs Version 1.

Output

- Digital Elevation Model
- Elevation Metadata

Step 5: Acquire GIS Datasets

Objective: Acquire supporting geospatial information describing infrastructure and environmental features.

Implementation

- Retrieve GIS layers for the selected AOI.
- Collect road networks, water bodies, rivers, and administrative boundaries.
- Validate the completeness of each layer.
- Convert datasets into a common spatial format.

Engineering Decision: OpenStreetMap (OSM) is selected as the primary GIS source because of its comprehensive geographic coverage, open licensing, and compatibility with GIS processing tools.

Output

- GIS Layers
- Feature Metadata

Step 6: Dataset Standardization

Objective: Ensure that all acquired datasets share a common spatial reference and are ready for integrated analysis.

Implementation

- Transform all datasets to a common CRS.
- Clip every dataset to the AOI boundary.
- Verify spatial alignment.
- Standardize file formats and metadata.
- Perform quality and completeness checks.

Engineering Decision: Standardizing all datasets before terrain analysis eliminates spatial inconsistencies and simplifies processing in subsequent modules.

Output

- Standardized Geospatial Dataset Package

Step 7: Dataset Organization

Objective: Organize the processed datasets using a standardized directory structure for efficient retrieval and management.

Implementation

- Create a dedicated directory for each AOI.
- Store Satellite, DEM, GIS, and Metadata separately.
- Maintain consistent naming conventions.
- Record dataset versions and acquisition dates.

Engineering Decision: A standardized directory structure improves maintainability, reproducibility, and simplifies data access for downstream modules.

Output

- Organized AOI Dataset Repository

Module 2: Candidate Terrain Identification

4.2.1 Overview

The **Candidate Terrain Identification** module is the first analytical stage of the Terrain Knowledge Generation Pipeline. Its objective is to identify **physically continuous terrain regions** that are suitable for further evaluation by eliminating areas occupied by permanent obstacles and unsuitable land features.

Rather than identifying helicopter landing locations, this module only determines **whether a terrain region is available for further analysis**. The generated candidate terrains serve as the fundamental building blocks for subsequent modules, where their geometric, environmental, and operational characteristics are extracted.

A **Candidate Terrain** is therefore defined as:

A contiguous region that remains after eliminating permanent obstacles and unsuitable terrain features from the Area of Interest. It represents terrain that is physically available for further evaluation but has not yet been assessed for helicopter operational suitability. The output of this module is a collection of **Terrain Objects**, each represented by its spatial boundary and unique identifier, which are forwarded to the Terrain Characterization module.

4.2.2 Inputs

The Candidate Terrain Identification module requires the standardized datasets generated by the Data Acquisition module.

Input	Purpose
Satellite Imagery	Identify terrain features and visible obstacles.
Digital Elevation Model (DEM)	Assist in distinguishing terrain structures and elevation characteristics.
GIS Layers	Detect roads, water bodies, buildings, and other mapped infrastructure.
AOI Metadata	Define the geographical boundary for analysis.

4.2.3 Methodology

The Candidate Terrain Identification module transforms raw geospatial information into **Terrain Objects** by progressively eliminating unsuitable regions and grouping the remaining terrain into contiguous candidate areas. Rather than directly searching for helicopter landing locations, this module identifies terrain that is **physically available for further evaluation**, while postponing operational suitability analysis to subsequent modules.

The methodology consists of the following stages:

Step 1: Terrain Classification

Objective: Classify every region within the Area of Interest into meaningful terrain categories that describe the physical nature of the land.

Implementation

- Process the satellite imagery covering the AOI.

- Classify terrain into predefined land-cover categories.
- Generate a terrain classification map where each pixel belongs to a specific terrain class.
- Typical terrain classes include:
 - Grassland
 - Bare Soil
 - Agricultural Land
 - Sparse Vegetation
 - Dense Vegetation / Forest
 - Water Bodies
 - Urban / Built-up Areas
 - Rocky Terrain

Engineering Decision: The system first determines **what each region represents** before deciding whether it should be considered. Separating terrain classification from obstacle removal creates a modular pipeline and allows future support for additional terrain categories without modifying downstream modules.

Output: A classified terrain map of the selected AOI.

Step 2: Permanent Obstacle Identification

Objective: Identify terrain regions that are permanently unsuitable for helicopter landing.

Implementation

Using the classified terrain map together with GIS information, identify regions occupied by:

- Buildings
- Dense forests
- Mountains / Cliff faces
- Large water bodies
- Urban infrastructure
- Other permanent obstacles

These regions are marked as **non-candidate terrain** and excluded from further processing.

Engineering Decision: Only **permanent obstacles** are removed during this stage. Dynamic objects such as vehicles, temporary structures, or weather-related obstacles are intentionally excluded since they are outside the scope of Version 1.

Output: Obstacle mask representing permanently unsuitable terrain.

Step 3: Candidate Terrain Extraction

Objective: Extract terrain regions that remain after removing permanent obstacles.

Implementation

- Remove obstacle pixels using the generated obstacle mask.
- Preserve only terrain classes that remain physically available.

- Ignore helicopter-specific requirements during this stage.
- Generate a binary candidate terrain map.

Engineering Decision

HELPSs intentionally avoids applying helicopter constraints during terrain extraction. This ensures that the generated terrain knowledge remains **helicopter-independent** and reusable across multiple aircraft platforms.

Output: Binary Candidate Terrain Map.

Step 4: Connected Terrain Formation

Objective: Identify continuous terrain regions by grouping neighboring candidate pixels into individual terrain objects.

Implementation

- Perform connected component analysis on the binary terrain map.
- Merge adjacent candidate pixels belonging to the same continuous land region.
- Assign a unique Terrain ID to every connected region.

Disconnected regions are treated as independent Terrain Objects.

Engineering Decision: Terrain is represented as **continuous connected regions** instead of individual pixels. This significantly reduces computational complexity and establishes the fundamental spatial entity used throughout HELPSs.

Output: Connected Terrain Objects.

Step 5: Terrain Polygon Generation

Objective: Convert connected terrain regions into geospatial polygons.

Implementation

- Extract the boundary of every connected terrain region.
- Generate polygon geometry representing the terrain extent.
- Store polygon coordinates together with the Terrain ID.

These polygons provide the spatial representation used throughout subsequent modules.

Engineering Decision Polygon representation preserves the complete geometry of each terrain object while enabling efficient spatial queries and geometric analysis.

Output: Terrain Object Polygons.

Step 6: Terrain Object Storage

Objective: Store the extracted Terrain Objects within the Terrain Knowledge Database.

Implementation

For each Terrain Object, store:

- Terrain ID
- Polygon Geometry
- Bounding Box
- AOI Reference
- Metadata

Only the geometric representation is stored at this stage. Terrain characteristics and operational properties are extracted in the following modules.

Engineering Decision: Separating **terrain geometry** from **terrain characteristics** keeps the database normalized and allows each subsequent module to incrementally enrich the Terrain Object without duplicating information.

Output: Terrain Object Repository.

Module 3: Terrain Characterization

This module is about **describing** every Terrain Object that we extracted.

4.3.1 Overview

The **Terrain Characterization** module enriches each Terrain Object with its physical, environmental, and geometric characteristics. Rather than determining whether a terrain is suitable for helicopter landing, this module focuses on **describing the terrain** by extracting attributes that will later support operational zone generation and helicopter-specific evaluation.

Each Terrain Object is analyzed independently, and its extracted characteristics are stored within the Terrain Knowledge Database. These characteristics remain **helicopter-independent**, ensuring that the same terrain information can be reused for multiple helicopter platforms without repeated analysis.

4.3.2 Inputs

Input	Purpose
Terrain Objects	Spatial boundaries generated by Module 2.
Satellite Imagery	Surface and land-cover analysis.
DEM	Terrain geometry and elevation analysis.
GIS Layers	Infrastructure and environmental information.

4.3.3 Methodology

The Terrain Characterization module extracts descriptive attributes from each Terrain Object. These attributes provide a comprehensive understanding of the terrain while remaining independent of any helicopter-specific operational constraints.

The methodology consists of the following stages:

Step 1: Geometric Characterization

Objective: Extract the fundamental geometric properties of each Terrain Object.

Implementation

For every Terrain Object, compute:

- Total Area
- Maximum Length
- Maximum Width

- Bounding Box
- Polygon Geometry

Engineering Decision: Geometric properties form the basis for subsequent analyses, including operational zone generation and landing footprint validation.

Output: Geometric characteristics of the Terrain Object.

Step 2: Surface Characterization

Objective: Determine the dominant surface type of each Terrain Object.

Implementation: Analyse the classified terrain map to determine the primary surface composition, such as:

- Bare Soil
- Grassland
- Agricultural Land
- Rocky Terrain
- Water Body

Store the dominant surface type together with its proportional coverage.

Engineering Decision: Surface characteristics influence helicopter landing feasibility and provide additional context during recommendation generation.

Output: Surface type information.

Step 3: Vegetation Analysis

Objective: Estimate the vegetation density present within each Terrain Object.

Implementation

- Analyze vegetation coverage using satellite imagery.
- Compute vegetation density for the Terrain Object.
- Store the overall vegetation percentage.

Engineering Decision: Rather than removing all vegetation, HELPSs records vegetation density as a terrain characteristic, allowing helicopter-specific evaluation in later stages.

Output: Vegetation density.

Step 4: Water Presence Analysis

Objective: Identify the presence of water bodies or water channels intersecting the Terrain Object.

Implementation

- Detect water bodies using GIS layers and satellite imagery.
- Determine whether water exists within the Terrain Object.
- Record the extent and location of water presence.

Engineering Decision: Water presence is stored as descriptive information rather than immediately rejecting the terrain, allowing later modules to determine its operational impact.

Output: Water presence information.

Step 5: Accessibility Analysis

Objective: Measure the accessibility of each Terrain Object.

Implementation

Compute:

- Distance to nearest road
- Road connectivity
- Nearby infrastructure

Engineering Decision: Accessibility is an operational characteristic useful for emergency response, rescue operations, and recommendation ranking.

Output: Accessibility attributes.

Step 6: Terrain Characteristic Storage

Objective: Store all extracted characteristics in the Terrain Knowledge Database.

Implementation: Associate the extracted attributes with the corresponding Terrain ID without modifying its geometry.

Store:

- Geometric Properties
- Surface Type
- Vegetation Density
- Water Presence
- Accessibility Attributes

Engineering Decision: Separating terrain geometry from terrain characteristics maintains database normalization and allows additional characteristics to be incorporated without altering the Terrain Object representation.

Output: Enriched Terrain Objects ready for Operational Zone Generation.

Module 4: Slope Analysis & Operational Zone Generation

4.4.1 Overview

The **Slope Analysis & Operational Zone Generation** module transforms each Terrain Object into one or more **Operational Zones** by analyzing terrain slope at multiple thresholds. Rather than assigning a single slope value to an entire terrain object, HELPSs preserves the internal slope variations by identifying contiguous regions that satisfy different slope limits.

This approach enables the system to maintain a **helicopter-independent representation** of terrain suitability. During the recommendation phase, helicopters with different operational slope limitations can retrieve the most appropriate Operational Zone without requiring repeated slope computation.

Each generated Operational Zone is stored independently together with its geometric properties, allowing the same Terrain Object to support multiple helicopter platforms and mission scenarios.

4.4.2 Inputs

Input	Purpose
Terrain Objects	Candidate terrain extracted from Module 2.
Terrain Characteristics	Geometric and environmental attributes from Module 3.
Digital Elevation Model (DEM)	Compute terrain slope.

4.4.3 Methodology

The methodology consists of the following stages:

Step 1 : Terrain Slope Computation



Step 2 : Multi-Threshold Slope Segmentation



Step 3 : Connected Operational Zone Extraction



Step 4 : Operational Zone Characterization



Step 5 : Pole of Inaccessibility Computation



Step 6 : Operational Zone Storage

Step 1: Terrain Slope Computation

Objective: Compute the terrain slope for every location within a Terrain Object using the Digital Elevation Model (DEM).

Implementation

- Extract DEM values corresponding to each Terrain Object.
- Compute the local terrain slope for every DEM cell.
- Generate a continuous slope raster representing the terrain.

Engineering Decision: Instead of assigning a single average slope to an entire Terrain Object, HELPSs computes **cell-level slope information** to preserve local terrain variations and enable more accurate operational zone extraction.

Output: Slope Raster.

Step 2: Multi-Threshold Slope Segmentation

Objective: Generate terrain regions satisfying different slope thresholds.

Implementation: Segment the slope raster using predefined thresholds:

- $\leq 0^\circ$
- $\leq 2^\circ$
- $\leq 4^\circ$

- $\leq 6^\circ$
- $\leq 8^\circ$
- $\leq 10^\circ$
- $\leq 12^\circ$
- $\leq 15^\circ$

Each threshold produces a binary slope mask representing terrain that satisfies the corresponding slope constraint.

Engineering Decision: HELPSs stores **multiple slope representations** instead of a single threshold. This allows helicopters with different allowable slope limits to reuse the same terrain knowledge without recomputing slope information.

Output: Multi-threshold slope masks.

Step 3: Connected Operational Zone Extraction

Objective: Extract contiguous regions that satisfy each slope threshold.

Implementation

- Perform connected component analysis on every slope mask.
- Separate disconnected regions into independent Operational Zones.
- Remove zones smaller than the minimum globally acceptable landing area.
- Split zones where internal water channels or excluded regions create disconnected land.

Engineering Decision: Each connected region is treated as an independent Operational Zone because different parts of the same Terrain Object may exhibit significantly different operational characteristics.

Output: Operational Zone candidates.

Step 4: Operational Zone Characterization

Objective: Extract geometric properties for every Operational Zone.

Implementation: For every Operational Zone, compute:

- Zone Area
- Maximum Length
- Maximum Width
- Polygon Geometry
- Bounding Box
- Associated Slope Threshold

These characteristics provide the geometric information required during helicopter-specific evaluation.

Engineering Decision: Each Operational Zone is independently characterized to support accurate landing footprint validation and helicopter-specific filtering.

Output: Characterized Operational Zones.

Step 5: Pole of Inaccessibility Computation

Objective: Determine the safest representative landing point within each Operational Zone.

Implementation

- Compute the **Pole of Inaccessibility** for every Operational Zone polygon.
- Measure its distance from the nearest polygon boundary.
- Store the point as the representative reference location for the zone.

Engineering Decision: Unlike a geometric centroid, the **Pole of Inaccessibility** lies at the location farthest from all boundaries, making it a more suitable reference for validating helicopter landing clearance. This improves operational safety, particularly for irregularly shaped zones.

Output: Representative Landing Point for every Operational Zone.

Step 6: Operational Zone Storage

Objective: Store all Operational Zones within the Terrain Knowledge Database.

Implementation: For every Operational Zone, store:

- Zone ID
- Parent Terrain ID
- Polygon Geometry
- Slope Threshold
- Area
- Length
- Width
- Bounding Box
- Pole of Inaccessibility
- Boundary Clearance Distance
- Maximum Inscribed Circle Radius (distance from that point to the nearest boundary)

Engineering Decision: Operational Zones are stored independently from Terrain Objects, allowing multiple zones with different slope characteristics to exist within the same terrain. This hierarchical representation enables efficient helicopter-specific evaluation while avoiding redundant terrain processing.

Output: Operational Zone Repository.

Module 5: Operational Zone Environmental Context Analysis

4.5.1 Overview

The **Operational Zone Environmental Context Analysis** module evaluates the surroundings of each Operational Zone to identify external environmental factors that may influence helicopter operations. While previous modules determine the geometric and topographical suitability of an Operational Zone, this module analyzes the **operational context** around the zone, including accessibility, surrounding obstacles, nearby water bodies, and approach feasibility.

The extracted environmental attributes are stored alongside each Operational Zone within the Terrain Knowledge Database and are later utilized during helicopter-specific evaluation and recommendation ranking.

4.5.2 Inputs

Input	Purpose
Operational Zones	Zones generated from Module 4.
Satellite Imagery	Analyze surrounding terrain features.
GIS Layers	Roads, water bodies, buildings, and infrastructure.
Terrain Characteristics	Environmental attributes of the parent Terrain Object.

4.5.3 Methodology

The methodology consists of the following stages:

Step 1: Accessibility Analysis

Objective: Evaluate how easily each Operational Zone can be accessed after an emergency landing.

Implementation

- Identify the nearest road or transportation network.
- Compute the distance from the Operational Zone to the nearest accessible road.
- Record accessibility information for rescue and recovery operations.

Engineering Decision: Accessibility is not used to reject an Operational Zone but provides valuable operational information during recommendation ranking and post-landing planning.

Output

- Distance to Nearest Road
- Accessibility Attributes

Step 2: Water Proximity Analysis

Objective: Determine the influence of nearby water bodies on helicopter operations.

Implementation

- Detect water bodies surrounding the Operational Zone.
- Measure the proximity of water features to the zone boundary.
- Identify water channels intersecting or immediately adjacent to the zone.

Engineering Decision: Rather than rejecting zones near water, HELPSs stores water proximity as contextual information that can influence recommendation scores based on operational requirements.

Output

- Water Proximity Information
- Water Interaction Attributes

Step 3: Surrounding Obstacle Analysis

Objective: Evaluate obstacles surrounding the Operational Zone that may affect helicopter approach and departure.

Implementation

- Define an analysis buffer around each Operational Zone.
- Analyze the surrounding region for trees, buildings, towers, cliffs, and other obstacles.
- Estimate the obstacle density and distribution within the buffer region.

Engineering Decision: Operational safety depends not only on the landing surface but also on the surrounding environment. Therefore, obstacle analysis extends beyond the zone boundary to provide a more realistic operational assessment.

Output

- Obstacle Density
- Surrounding Obstacle Information

Step 4: Approach Corridor Analysis

Objective: Estimate the openness of the surrounding terrain to support safe helicopter approach and departure.

Implementation

- Analyze the terrain surrounding the Operational Zone.
- Evaluate the availability of unobstructed space around the zone.
- Compute an **Approach Corridor Score** representing the overall openness of the surrounding environment.

Engineering Decision: Instead of computing an exact flight approach path, HELPSs estimates the openness of the surrounding terrain. This provides a practical and computationally efficient approximation of approach feasibility while remaining independent of helicopter-specific flight dynamics.

Output

- Approach Corridor Score

Step 5: Environmental Context Storage

Objective: Associate the extracted environmental context with each Operational Zone.

Implementation: Store the following attributes:

- Distance to Nearest Road
- Water Proximity
- Obstacle Density
- Approach Corridor Score
- Environmental Metadata

Engineering Decision: Separating environmental context from geometric and topographical characteristics maintains a modular database structure while enabling helicopter-specific recommendation algorithms to selectively utilize contextual information.

Output: Operational Zones enriched with environmental context.

Chapter 5: Terrain Knowledge Database

5.1 Overview

The Terrain Knowledge Database serves as the centralized repository for all information generated during the Terrain Knowledge Generation Pipeline. We also actively hold all the restricted areas coordinates and exclude it from location candidates. Rather than storing raw geospatial datasets, the database stores **processed terrain knowledge**, allowing the Helicopter Landing Recommendation Pipeline to perform efficient spatial queries without repeating computationally expensive terrain analysis.

The database follows a **hierarchical structure**, where each Terrain Object represents a physically continuous region and each Terrain Object contains one or more Operational Zones generated from multi-threshold slope analysis. This organization eliminates redundant computation while enabling multiple helicopter platforms to evaluate the same terrain using their respective operational constraints.

5.2 Inputs

Input	Source
Terrain Objects	Module 2
Terrain Characteristics	Module 3
Operational Zones	Module 4
Zone Geometry & Metadata	Module 4

5.3 Methodology

The Terrain Knowledge Database is populated incrementally as each module of the Terrain Knowledge Generation Pipeline completes. Instead of duplicating information, HELPSs follows a **normalized hierarchical storage model**, where every module enriches previously generated entities.

The methodology consists of the following stages:

Step 1: Store Terrain Objects

For every Terrain Object, store:

- Terrain ID
- Polygon Geometry
- Bounding Box
- AOI Reference

Step 2: Store Terrain Characteristics

Associate the following attributes with the corresponding Terrain Object:

- Area
- Length
- Width
- Surface Type

- Vegetation Density
- Water Presence
- Accessibility Attributes

Step 3: Store Operational Zones

Each Terrain Object may contain multiple Operational Zones corresponding to different slope thresholds.

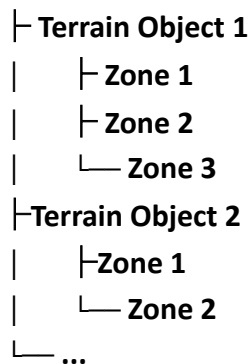
For every Operational Zone, store:

- Zone ID
- Parent Terrain ID
- Polygon Geometry
- Slope Threshold
- Area
- Length
- Width
- Pole of Inaccessibility
- Maximum Inscribed Circle Radius

Step 4: Establish Relationships

Maintain hierarchical relationships:

AOI



This structure enables efficient retrieval of all zones belonging to a terrain object while preserving normalization.

Step 5: Indexing & Query Optimization

Create indexes on:

- AOI ID
- Terrain ID
- Zone ID
- Spatial Geometry
- Slope Threshold

These indexes improve retrieval performance during the Helicopter Landing Recommendation Pipeline.

4.5.4 Outputs

The output of this module is a fully populated **Terrain Knowledge Database** containing structured, helicopter-independent terrain information that can be queried efficiently during emergency landing recommendation.

4.5.5 Validation & Limitations

Database validation ensures referential integrity between Terrain Objects and Operational Zones, verifies the consistency of stored geometries, and confirms the correctness of spatial indexing. Since the database stores processed terrain knowledge, any changes in the underlying geospatial datasets require regeneration of the affected terrain information.

Chapter 6: Helicopter Landing Recommendation Pipeline

6.1 Overview

The **Helicopter Landing Recommendation Pipeline** utilizes the precomputed Terrain Knowledge Database to identify and recommend suitable emergency landing locations for a specific helicopter. Unlike the Terrain Knowledge Generation Pipeline, which performs computationally intensive terrain analysis, this pipeline focuses on **helicopter-specific evaluation** by considering the aircraft's operational limitations, mission requirements, and prevailing environmental conditions.

Rather than processing raw satellite imagery or Digital Elevation Models (DEM), the system retrieves precomputed **Operational Zones** and progressively evaluates them through a sequence of operational filters and ranking mechanisms. The evaluation ensures that only physically feasible and operationally safe landing locations are considered before generating a ranked set of recommendations.

The recommendation process follows seven sequential modules:

- **Module 1:** Helicopter Input Processing
- **Module 2:** Nearby Operational Zone Retrieval
- **Module 3:** Weather & Operational Feasibility Assessment
- **Module 4:** Operational Constraint Matching
- **Module 5:** Landing Footprint Validation
- **Module 6:** Mission-Aware Ranking Engine
- **Module 7:** Recommendation Generation

6.2 Inputs

Input	Purpose
Helicopter Current Coordinates	Retrieve nearby Operational Zones.
Helicopter Model	Identify helicopter-specific operational characteristics.
Helicopter Operational Specifications	Retrieve landing limitations such as allowable slope, landing footprint, safety clearances, and operational constraints.
Mission Context	Prioritize recommendations according to mission objectives.
Current Weather & Wind Information	Assess operational feasibility and influence recommendation ranking.
Terrain Knowledge Database	Retrieve Operational Zones and their associated terrain knowledge.

6.3 Methodology

The Helicopter Landing Recommendation Pipeline evaluates precomputed Operational Zones through a sequence of helicopter-specific validation stages. Each stage progressively reduces the candidate set by eliminating operationally infeasible zones and enriching the

remaining candidates with additional evaluation metrics. Only Operational Zones that satisfy all mandatory operational constraints are forwarded to the Mission-Aware Ranking Engine, where they are prioritized according to mission objectives before generating the final recommendations.

The methodology consists of the following modules:

1. Helicopter Input Processing
2. Nearby Operational Zone Retrieval
3. Weather & Operational Feasibility Assessment
4. Operational Constraint Matching
5. Landing Footprint Validation
6. Mission-Aware Ranking Engine
7. Recommendation Generation

Module 1: Helicopter Input Processing

Objective: Collect all helicopter-specific information required for evaluating Operational Zones.

Implementation

- Acquire the helicopter's current GPS coordinates.
- Identify the helicopter model.
- Retrieve helicopter-specific operational specifications.
- Load mission objectives and operational priorities.
- Obtain current weather and wind information.
- Construct the Helicopter Operational Profile used throughout the recommendation process.

Engineering Decision: Separating helicopter-specific information from terrain knowledge enables HELPSs to support multiple helicopter platforms without modifying the Terrain Knowledge Database.

Output: Helicopter Operational Profile.

Module 2: Nearby Operational Zone Retrieval

Objective: Retrieve all Operational Zones located within the helicopter's operational search region from the Terrain Knowledge Database. This reduces the search space by limiting the evaluation to geographically relevant landing candidates.

Implementation

- Use the helicopter's current GPS coordinates as the reference point.
- Define a configurable operational search radius around the helicopter.
- Perform a spatial query on the Terrain Knowledge Database to retrieve all Operational Zones within the search region.
- Compute the distance between the helicopter's current location and the **Pole of Inaccessibility** of each retrieved Operational Zone.

- Retrieve the associated terrain characteristics and environmental context for every Operational Zone.
- Construct the initial candidate set for subsequent evaluation modules.

Engineering Decision: HELPSs retrieves **Operational Zones** instead of raw terrain because they already represent geometrically valid and slope-compliant landing regions generated during the Terrain Knowledge Generation Pipeline. Computing the helicopter-to-zone distance during retrieval avoids repeated calculations in later stages and provides an important parameter for recommendation ranking.

Output

Candidate Operational Zone Set containing:

- Operational Zone ID
- Parent Terrain ID
- Pole of Inaccessibility
- Distance from Helicopter
- Terrain Characteristics
- Environmental Context

Module 3: Weather & Operational Feasibility Assessment

Objective: Determine whether the current environmental conditions permit safe helicopter operations before performing computationally intensive landing evaluations and save the weather conditions to reuse of other nearby locations .

Implementation

- Retrieve the current weather and wind conditions for the operational area.
- Compare the environmental conditions against the helicopter's operational weather limits.
- Evaluate parameters such as:
 - Wind Speed
 - Wind Direction
 - Visibility
 - Rainfall Intensity
 - Temperature (*if applicable*)
 - Other helicopter-specific operational weather limits
- If any mandatory operational limit is violated, terminate the recommendation process.
- Otherwise, associate the environmental information with every candidate Operational Zone for use during ranking.

Engineering Decision: Weather serves two distinct purposes within HELPSs. It first acts as a **hard operational feasibility filter**, preventing recommendations under conditions where helicopter operations are unsafe or prohibited. If the weather satisfies all mandatory operational limits, it is subsequently retained as an environmental factor that can influence recommendation ranking without rejecting otherwise feasible landing locations.

Output

- Operationally Feasible Candidate Operational Zones
- Weather Assessment Report

Module 4: Operational Constraint Matching

Objective: Evaluate each candidate Operational Zone against the helicopter's operational limitations to eliminate zones that are physically unsuitable for landing. This module ensures that only Operational Zones satisfying the helicopter's mandatory operational constraints are forwarded for detailed landing validation.

Implementation

- Retrieve the helicopter's operational specifications from the Helicopter Operational Profile.
- Compare the helicopter's maximum allowable landing slope with the stored slope threshold of each Operational Zone.
- Verify that the Operational Zone provides at least the minimum landing area required by the helicopter.
- Compare the **Maximum Inscribed Circle Radius** of the Operational Zone against the helicopter's required landing clearance radius.
- Reject Operational Zones that fail any mandatory operational constraint.
- Forward the remaining Operational Zones to the Landing Footprint Validation module.

Engineering Decision: HELPSs performs operational constraint matching using **precomputed Operational Zones** rather than recomputing terrain characteristics during every query. Since Operational Zones are already classified according to multiple slope thresholds, helicopters with different operational limitations can efficiently retrieve compatible zones without repeating computationally expensive terrain analysis.

Output: Operationally Compatible Operational Zones.

Module 5: Landing Footprint Validation

Objective: Verify that the helicopter can be safely accommodated within the selected Operational Zone while maintaining the required safety clearance from surrounding boundaries.

Implementation

- Use the **Pole of Inaccessibility** as the initial candidate landing point.
- Construct the helicopter's landing footprint using its required landing clearance radius.
- Position the landing footprint at the Pole of Inaccessibility.
- Verify that the landing footprint lies entirely within the Operational Zone polygon.
- Ensure that the required safety buffer is maintained from all zone boundaries.
- Reject Operational Zones that cannot safely accommodate the helicopter's landing footprint.

Engineering Decision: The Pole of Inaccessibility represents the point farthest from the Operational Zone boundary, making it the safest initial landing reference. Validating the

helicopter's landing footprint around this point provides a practical and computationally efficient method for ensuring sufficient landing clearance while minimizing the risk of rotor or airframe boundary conflicts.

Output: Landing Validated Operational Zones.

Module 6: Mission-Aware Ranking Engine

Objective: Evaluate all operationally feasible landing locations and rank them according to mission requirements, operational priorities, and environmental context to generate the most suitable emergency landing recommendations.

Implementation

- Retrieve the validated Operational Zones from the previous module.
- Extract the static terrain and environmental characteristics associated with each Operational Zone.
- Incorporate the current weather and wind conditions.
- Identify the active mission context and corresponding operational priorities.
- Compute a configurable Suitability Score for each Operational Zone.
- Rank the Operational Zones according to the computed suitability score.
- Forward the ranked list to the Recommendation Generation module.

The suitability score is conceptually represented as:

$$\text{Suitability Score} = f(\text{Static Terrain Features}, \text{Dynamic Weather Conditions}, \text{Mission Context})$$

Where:

- Static Terrain Features include:
 - Distance from Helicopter
 - Distance to Nearest Road
 - Accessibility
 - Surface Type
 - Vegetation Density
 - Water Proximity
 - Approach Corridor Score
- Dynamic Weather Conditions include:
 - Wind Speed & Direction
 - Visibility
 - Rainfall
 - Other operational weather parameters
- Mission Context adapts the ranking strategy according to the current mission objective. Examples include:
 - **Medical Evacuation** → prioritize road accessibility and proximity to populated areas.
 - **Engine Failure** → prioritize immediate landing safety.

- **Fuel Emergency** → prioritize proximity to the helicopter.

Engineering Decision: Unlike previous modules, the Mission-Aware Ranking Engine **does not reject Operational Zones**. Instead, it assigns a relative suitability score to each operationally feasible location. This separation between **mandatory operational validation** and **preference-based ranking** improves explainability and allows the ranking strategy to be adapted for different mission profiles without modifying the core recommendation pipeline.

Output: Ranked Operational Zones with Suitability Scores.

Module 7: Recommendation Generation

Objective: Generate the final set of landing recommendations and present them in an interpretable format for pilot decision support.

Implementation

- Select the highest-ranked Operational Zones.
- Generate the Top-K landing recommendations.
- Associate each recommendation with its supporting information.
- Present the recommended landing point using the **Pole of Inaccessibility** coordinates.
- Include the suitability score and the factors contributing to the recommendation.
- Display all recommendations in descending order of suitability.

Each recommendation includes:

- Operational Zone ID
- Landing Coordinates
- Distance from Helicopter
- Suitability Score
- Terrain Characteristics
- Environmental Context
- Recommendation Explanation

Engineering Decision: HELPSs is designed as a **Decision Support System**, not an autonomous decision-making system. Rather than selecting a single landing location, the system presents multiple explainable recommendations, enabling the pilot to make the final operational decision based on real-time situational awareness.

Output: Ranked and Explainable Emergency Landing Recommendations.

6.4 Outputs

The Helicopter Landing Recommendation Pipeline produces a prioritized list of operationally feasible emergency landing locations. Each recommendation includes the validated landing point, operational suitability score, terrain characteristics, environmental context, and supporting information that explains why the location was recommended. The output is intended to assist the pilot in making informed emergency landing decisions while retaining full pilot authority over the final selection.

6.5 Validation & Limitations

The recommendation pipeline is validated by verifying that every recommended Operational Zone satisfies all mandatory helicopter operational constraints before ranking. The correctness of the recommendations depends on the accuracy of the Terrain Knowledge Database, helicopter operational specifications, and environmental data used during evaluation.

Since HELPSs is a **decision support system**, it does not account for dynamic obstacles, rapidly changing environmental conditions, or pilot judgment during the final approach. The recommendations should therefore be treated as operational guidance rather than autonomous landing commands.

Chapter 7: Design Decisions and Engineering Components

7.1 Terrain Representation

One of the fundamental design decisions in HELPSs is representing the terrain as **Terrain Objects** instead of processing raw satellite imagery or Digital Elevation Models during every recommendation request. Each Terrain Object represents a **physically continuous candidate terrain** identified during the Terrain Knowledge Generation Pipeline and serves as the primary spatial entity within the Terrain Knowledge Database.

Terrain Objects are stored as **polygon geometries**, enabling efficient spatial queries, geometric analysis, and database storage. This representation eliminates repeated terrain processing, improves computational efficiency, and allows the same terrain knowledge to be reused across multiple helicopter platforms and mission scenarios.

Engineering Decision: Representing terrain as reusable polygon-based Terrain Objects separates **terrain knowledge generation** from **helicopter recommendation**, resulting in a modular, scalable, and computationally efficient system.

7.2 Connected Terrain Extraction

Identifying isolated land pixels is insufficient for helicopter landing assessment, as helicopters require **continuous usable terrain** rather than scattered land patches. Therefore, HELPSs groups adjacent candidate terrain pixels into **connected regions**, where each region represents a single Terrain Object.

This approach preserves the actual spatial continuity of the terrain, enables accurate geometric analysis, and forms the foundation for subsequent processes such as terrain characterization, slope analysis, and operational zone generation.

Engineering Decision: HELPSs represents terrain as **connected regions** instead of individual pixels to ensure that every Terrain Object corresponds to a physically continuous land area suitable for further operational evaluation. This significantly reduces computational complexity while providing a meaningful representation of the terrain.

7.3 Multi-Threshold Slope Representation

Traditional terrain analysis often represents an entire region using a **single average slope**, which can overlook significant local variations within the terrain. HELPSs instead adopts a **multi-threshold slope representation**, where each Terrain Object is analyzed across multiple slope thresholds (e.g., $\leq 0^\circ$, $\leq 2^\circ$, $\leq 4^\circ$, $\leq 6^\circ$, $\leq 8^\circ$, $\leq 10^\circ$, $\leq 12^\circ$, and $\leq 15^\circ$). This preserves the internal slope distribution and identifies all operationally feasible regions within the terrain. By storing Operational Zones corresponding to each slope threshold, the same terrain knowledge can support helicopters with different allowable landing slope limits without recomputing terrain characteristics. This enables efficient helicopter-specific evaluation while maintaining a single reusable Terrain Knowledge Database.

Engineering Decision: HELPSs stores **multiple slope-based Operational Zones** rather than a single terrain slope value. This preserves terrain variability, supports multiple helicopter platforms, and eliminates repeated slope analysis during the recommendation phase.

7.4 Operational Zone Representation

A single Terrain Object may contain multiple regions with different slope characteristics, making the entire terrain unsuitable to be represented by a single landing location. HELPSs therefore divides each Terrain Object into one or more **Operational Zones**, where each zone represents a contiguous region satisfying a specific slope threshold.

Operational Zones are stored independently while maintaining their relationship with the parent Terrain Object. This hierarchical representation preserves terrain variability, avoids redundant terrain analysis, and enables efficient helicopter-specific retrieval during the recommendation process.

Engineering Decision: HELPSs adopts a **hierarchical Terrain Object–Operational Zone model** instead of representing each terrain with a single landing point. This allows multiple helicopters with different operational capabilities to evaluate the same terrain independently while maintaining a normalized and scalable Terrain Knowledge Database.

7.5 Landing Reference Point Selection

Selecting the geometric **centroid** of an Operational Zone does not always provide a safe landing location, particularly for irregularly shaped regions where the centroid may lie close to the boundary or even outside the usable area. To overcome this limitation, HELPSs uses the **Pole of Inaccessibility** as the representative landing point, as it corresponds to the location farthest from all surrounding boundaries.

This point also provides the **Maximum Inscribed Circle Radius**, representing the largest circular landing footprint that can fit entirely within the Operational Zone. Together, these measures improve landing safety and simplify subsequent landing footprint validation.

Engineering Decision: HELPSs adopts the **Pole of Inaccessibility** instead of the centroid because it maximizes boundary clearance and provides a safer, more reliable reference point for helicopter landing evaluation.

7.6 Landing Footprint Validation

A location with sufficient area is not necessarily suitable for helicopter landing. The helicopter's landing footprint, including the required safety clearance around the aircraft, must fit completely within the Operational Zone without intersecting its boundaries or internal exclusion regions.

HELPSs validates the landing footprint by placing the helicopter's required clearance circle at the Pole of Inaccessibility and verifying that it remains entirely within the usable Operational Zone. This geometric validation ensures that every recommended landing

location can physically accommodate the helicopter while maintaining the prescribed safety margins.

Engineering Decision: HELPSs performs **geometric landing footprint validation** after operational constraint matching to ensure that only physically feasible landing locations are recommended, improving both operational safety and recommendation reliability.

7.7 Environmental Context Representation

Terrain geometry alone is insufficient to determine the operational suitability of a landing location. Factors such as surrounding obstacles, accessibility, nearby water bodies, and available approach space significantly influence the practicality and safety of helicopter operations. HELPSs therefore associates each Operational Zone with an **Environmental Context**, providing additional information about its surroundings without modifying the underlying terrain representation.

This separation enables environmental attributes to influence recommendation quality while preserving the helicopter-independent nature of the Terrain Knowledge Database.

Engineering Decision: HELPSs stores environmental information as a separate contextual layer rather than embedding it within terrain geometry. This improves modularity, simplifies future extensions, and enables different recommendation strategies without modifying the terrain representation.

7.8 Configurable Recommendation Framework

Different emergency scenarios require different landing priorities. For example, a medical evacuation may prioritize accessibility, whereas an engine failure may prioritize the safest immediately reachable landing location. To accommodate these varying operational requirements, HELPSs employs a **configurable recommendation framework** instead of a fixed scoring model.

The final suitability score is computed by combining **Static Terrain Features**, **Dynamic Weather Conditions**, and **Mission Context**, allowing the recommendation strategy to adapt without altering the underlying Terrain Knowledge Database.

Engineering Decision: HELPSs separates the recommendation strategy from terrain knowledge generation by adopting a configurable scoring framework. This enables future integration of alternative ranking techniques, mission-specific priorities, and optimization algorithms without requiring architectural changes.

7.9 Separation of Knowledge Generation and Recommendation

One of the key architectural decisions in HELPSs is the separation of **Terrain Knowledge Generation** from the **Helicopter Landing Recommendation Pipeline**. Terrain analysis is

performed once to generate reusable terrain knowledge, which is stored within the Terrain Knowledge Database. During recommendation, the system queries this precomputed knowledge instead of repeating computationally expensive terrain analysis.

This separation improves computational efficiency, reduces response time, supports multiple helicopter platforms, and enables the Terrain Knowledge Database to be updated independently of the recommendation engine.

Engineering Decision: HELPSs adopts a **two-stage architecture** consisting of offline terrain knowledge generation and online helicopter recommendation. This modular design improves scalability, reusability, maintainability, and overall system performance.

7.10 Chapter Summary

The engineering decisions presented in this chapter define the core architectural principles of HELPSs. Concepts such as **Terrain Objects**, **Operational Zones**, **multi-threshold slope representation**, **Pole of Inaccessibility**, **environmental context modeling**, and the **separation of terrain knowledge generation from recommendation** collectively improve the system's modularity, computational efficiency, scalability, and adaptability. These design choices establish a reusable terrain knowledge framework capable of supporting diverse helicopter platforms, mission profiles, and future enhancements without requiring fundamental architectural changes.

Chapter 8: System Development Roadmap

8.1 Development Philosophy

The development of HELPSs follows a **modular and incremental approach**, where the system is constructed through a sequence of independent yet interconnected phases. Each phase produces a well-defined deliverable that serves as the foundation for the subsequent phase, enabling progressive system development, easier debugging, and independent validation of individual components.

Rather than building the complete system as a single application, HELPSs separates **terrain knowledge generation** from **helicopter recommendation**, allowing computationally intensive terrain analysis to be performed once and reused across multiple recommendation requests. This phased development strategy improves maintainability, scalability, and simplifies future system enhancements.

8.2 Phase 1: Geospatial Data Infrastructure

Objective

Establish the geospatial data acquisition and preprocessing pipeline required for terrain analysis.

Activities

- Implement Area of Interest (AOI) selection.
- Acquire satellite imagery, DEM, GIS datasets, and Restricted Area Layers.
- Standardize coordinate reference systems.
- Align dataset resolutions and spatial extents.
- Organize datasets into the standardized project structure.

Deliverable

A standardized geospatial dataset package containing all required datasets for a selected Area of Interest.

8.3 Phase 2: Candidate Terrain Generation

Objective

Identify physically continuous terrain suitable for further analysis.

Activities

- Perform terrain classification.
- Detect permanent obstacles.
- Exclude restricted operational areas.
- Perform connected terrain extraction.
- Generate Terrain Objects.

Deliverable

A collection of Terrain Objects representing candidate terrain within the selected Area of Interest.

8.4 Phase 3: Terrain Characterization

Objective

Extract descriptive characteristics for every Terrain Object.

Activities

- Compute geometric properties.
- Determine surface characteristics.
- Analyze vegetation density.
- Detect water presence.
- Compute accessibility attributes.

Deliverable

Terrain Objects enriched with geometric, environmental, and accessibility characteristics.

8.5 Phase 4: Operational Zone Generation

Objective

Generate helicopter-independent Operational Zones from Terrain Objects.

Activities

- Compute terrain slope from DEM.
- Generate multi-threshold slope maps.
- Extract connected Operational Zones.
- Generate polygon geometries.
- Compute the Pole of Inaccessibility.
- Compute the Maximum Inscribed Circle Radius.
- Associate Environmental Context with every Operational Zone.

Deliverable

Operational Zones representing physically feasible landing regions together with their geometric and environmental properties.

8.6 Phase 5: Terrain Knowledge Database

Objective

Organize all generated terrain knowledge into a reusable database.

Activities

- Design the Terrain Knowledge Database.
- Store Terrain Objects and their characteristics.
- Store Operational Zones.
- Store Environmental Context.
- Establish relationships between all entities.

Deliverable

A fully populated Terrain Knowledge Database capable of supporting helicopter recommendation queries.

8.7 Phase 6: Helicopter Recommendation Engine

Objective

Develop the recommendation pipeline that evaluates Operational Zones for specific helicopter missions.

Activities

- Develop helicopter input processing.
- Implement Operational Zone retrieval.
- Integrate weather and operational feasibility assessment.
- Implement operational constraint matching.
- Perform landing footprint validation.
- Develop the Mission-Aware Ranking Engine.
- Generate explainable landing recommendations.

Deliverable

A complete Helicopter Landing Recommendation Engine capable of producing ranked emergency landing recommendations.

8.8 Phase 7: User Interface and System Integration**Objective**

Integrate all system components into a unified application.

Activities

- Develop the local user interface.
- Integrate the Terrain Knowledge Database.
- Connect the recommendation engine.
- Visualize Operational Zones and recommendations on maps.
- Display supporting recommendation information and suitability scores.

Deliverable

A fully integrated HELPSs prototype providing end-to-end emergency landing recommendations.

8.9 Phase 8: System Optimization and Enhancement**Objective**

Improve the performance, scalability, and usability of the integrated system.

Activities

- Optimize spatial database queries.
- Improve recommendation response time.
- Optimize data storage and retrieval.
- Enhance ranking strategies.
- Prepare the architecture for future feature integration.

Deliverable

An optimized, scalable, and deployment-ready HELPSs framework.

8.10 Chapter Summary

The development of HELPSs follows a structured, phase-wise implementation strategy in which each phase produces a well-defined deliverable that serves as the foundation for the next stage. This modular approach enables incremental development, simplifies system integration, and supports scalable expansion as new functionalities and helicopter platforms are introduced.

Phase	Key Outcome / Deliverable
Phase 1 – Geospatial Data Infrastructure	Standardized geospatial datasets prepared for analysis.
Phase 2 – Candidate Terrain Generation	Terrain Objects representing physically continuous candidate terrain.
Phase 3 – Terrain Characterization	Terrain Objects enriched with geometric and environmental attributes.
Phase 4 – Operational Zone Generation	Operational Zones generated with slope, geometry, and environmental context.
Phase 5 – Terrain Knowledge Database	Structured Terrain Knowledge Database containing reusable terrain knowledge.
Phase 6 – Helicopter Recommendation Engine	Helicopter-specific recommendation pipeline capable of generating ranked landing candidates.
Phase 7 – User Interface & System Integration	Fully integrated HELPSs prototype with visualization and recommendation capabilities.

Phase 8 – System Optimization & Enhancement	Optimized, scalable, and deployment-ready HELPSs framework.
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Key Takeaways

- **Modular development** enables each subsystem to be developed and validated independently.
- **Terrain knowledge is generated once** and reused across multiple recommendation requests.
- **Each phase produces a measurable deliverable**, ensuring systematic project progress.
- The roadmap provides a **clear implementation pathway** from raw geospatial data to a complete helicopter emergency landing recommendation system.
- The phased architecture supports **future scalability, maintainability, and integration of advanced capabilities** without major architectural changes.